SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Code of Conduct

I Karthik.C(1925Roo) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Karthik

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Cloud Computing and Big Data Analytics

1. Compare Type-1 and Type-2 Hypervisors for a cloud Datacenter.

A hypervisor is software that creates and manages Virtual machines (Vms). There are two main :- type I (Bare-metal) and Type-2 (Hosted).

Features	Type-I (Hypervisor)	Type-2 Hypervisor
Installation	Directly on hardware	On top of an operating system
Performance	Very high	moderate
Security	Strong	Lower
Reliability	Excellent	Depends on host OS
Examples	VMware, ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation.

Analysis:

→ Performance: Type-1 offers better CPU, memory, and storage Performance because it runs directly on hardware.

→ Security: It has a smaller attack surface and provides better isolation b/w virtual machines.

→ Manageability: Supports Centralized management, live migration high availability, and disaster recovery.

2. Virtualization Architecture for Workload Isolation.

Virtualization enables multiple Organizations to share the same physical server while keeping their workloads isolated.

How Isolation is Achieved:

→ Each department runs in a seperate virtual machine.

→ Every VM has it own operating system, memory, CPU and storage.

→ The hypervisor prevents one VM for accessing another VM's resources.

Major Attack Surfaces:

1. Hypervisor Attack: If the Hypervisor is Compromised, all virtual machines are at risks.

2. VM Escape Attack: A malicious application inside one VM may attempt to access the host or other VMs.

Security Measures:-

-> Strong access control

-> Network segmentation

-> Encryption

-> Regular security patches

-> Multi-factor authentication (MFA).

3. VM Host Utilization Analysis

CPU Usage = 92%

Memory usage = 88%

Storage I/O latency = 35ms

VM Boot Time = 170 seconds.

Analysis:-

CPU (92%)

-> Indicates CPU overload

-> Application may become slow.

Memory (88%)

-> High memory usage may cause swapping and reduce Performance.

Storage I/O Latency (35ms)

-> High latency means slow disk access

-> Affects application response time.

VM Boot time (170s)

-> Boot time is longer than normal due to heavy resources usage.

Corrective Actions:

→ Upgrade CPU resources.

→ Increase RAM

→ Use SSD or NVMe storage

→ Balance workloads using VM migration.

→ Monitor resources continuously.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter.

Traditional Datacenter

→ Hardware configuration is mostly manual

→ Scaling takes more time

→ Limited automation.

Software Defined Datacenter

→ All infrastructure is controlled through software.

→ Supports automation and centralized management.

Advantages of SDDC

1. Compute Virtualization

2. Storage Virtualization

3. Network Virtualization.

Benefits:-

→ Faster deployment

→ Better scalability

→ lower operational cost

→ Improved resource utilization.

5 Multi-Cloud Federation Identity Problem

Users Experience login failures because different cloud Provider use different identity systems.

Causes :

-> Different authentication methods
-> Different user databases.
-> Lack of trust b/w providers.

Secure Identity Design

-> Implement single sign-on (SSO)
-> Use Identity Federation
-> Apply Multi-Factor Authentication (MFA)
-> Encrypt Identity information.
-> Use Role-Based Access control (RBAC)
-> Enable secure token-based Authentication.

Privacy Protection.

-> Share only necessary user information.
-> Encrpt data during storage and transmission.
-> Maintain audit logs for monitoring.